(certain) RESEARCH STORY

The real question is which customer job Digiroom can help finish.

A visual research story for the Auto2000 case

Existing tools already do useful work. The open opportunity is more specific: a decision, record or handoff that the customer cannot yet use with confidence.

Start with value

Completed jobs and contribution

Test the evidence

States, samples and source limits

Keep choices open

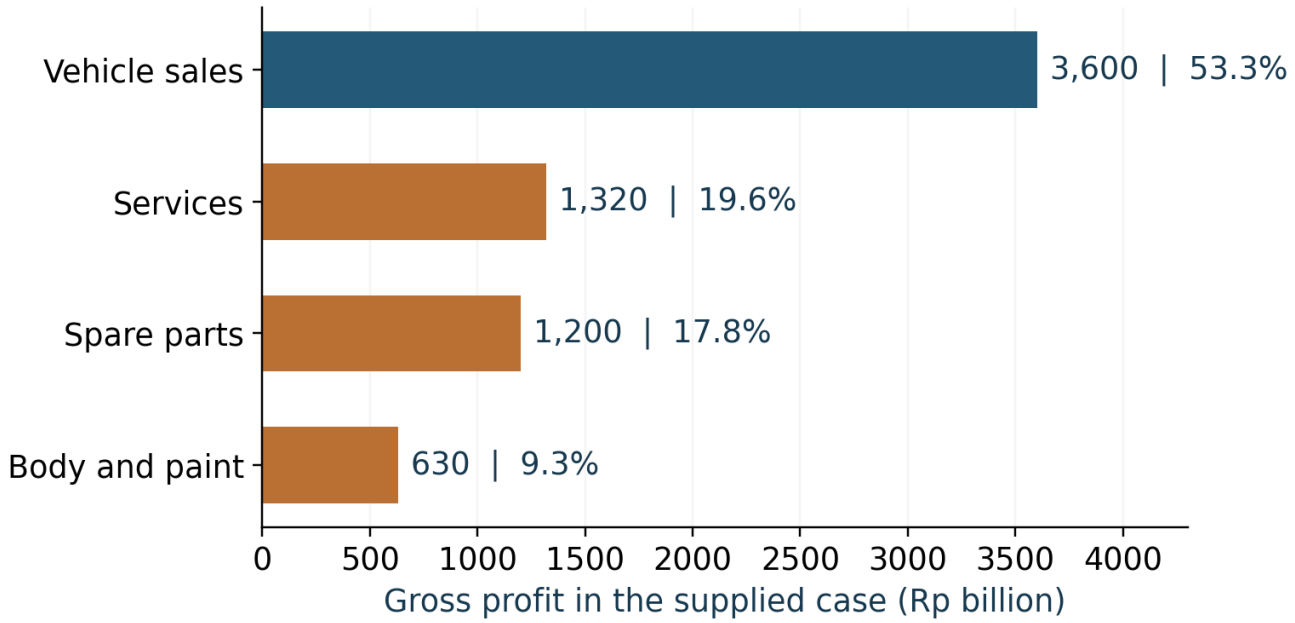
27 concepts; no selected winner

(certain) marks captured states or reproduced calculations. (guessing) marks an untested mechanism. Charts preserve those distinctions. Full technical work and raw data remain in the ZIP.

(certain) SUPPLIED CASE EXHIBIT

Aftersales matters more than its revenue share suggests.

The supplied case assigns Rp3,150 billion of gross profit to services, spare parts and body and paint, against Rp9,000 billion of revenue. Vehicle sales remain the largest individual profit pool. This makes the economics of completed ownership jobs material even when they generate fewer app sessions. The four lines belong to one case P&L, not four customer samples. A programme still needs incremental contribution after its own costs.



A useful ownership job can matter commercially even when it produces few app sessions.

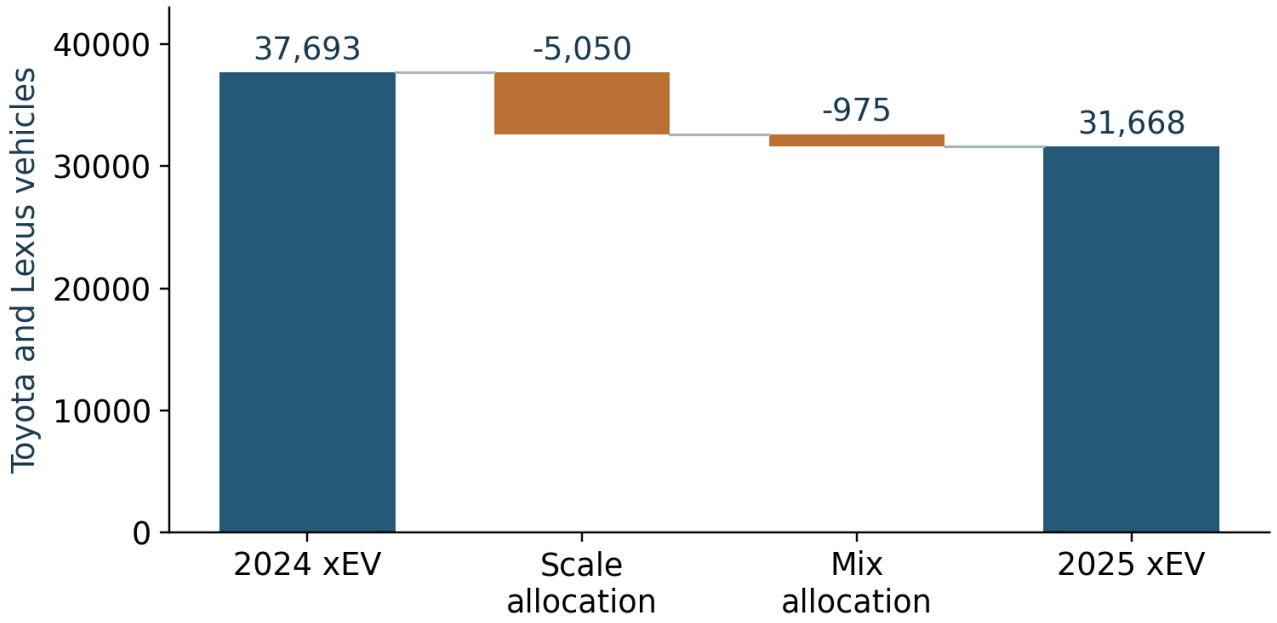
Limit: GP differs from operating profit and programme cash flow. Line-average margins do not identify EV-specific or per-job contribution.

Source: supplied IBSC casebook, Exhibit 3, printed/PDF p10. Business-line case assumptions, not audited Auto2000 results.

(certain) PUBLISHED / DERIVED

Most of the xEV decline matches a smaller total portfolio.

Astra reports 37,693 Toyota and Lexus xEV sales in 2024 and 31,668 in 2025. The symmetric accounting bridge assigns about 5,050 fewer units to the total-portfolio scale term and 975 to its xEV mix term. Averaging both calculation orders removes an arbitrary ordering choice. These terms reconcile the observed change; they do not measure the effects of national demand, Digiroom or any proposed solution.



Separate broad sales scale, vehicle mix and dealer conversion before blaming an app.

Limit: Do not label the scale term a national-market effect. Do not label the mix term a Digiroom effect.

Source: Astra AR2025, printed p140/PDF p142. Toyota AND Lexus domestic new sales. Shapley terms are accounting allocations, not causes or Auto2000 volume.

(certain, conditional) MECE ACCOUNTING FRAMEWORK

Measure completed jobs and their contribution

Start with eligible jobs in a defined vehicle cohort and period. Then ask how many Auto2000 completes, and what revenue minus recognized direct cost each completed job earns. The fourth check covers extra operating cost, displacement and retained risk. Compare totals across all channels against a defensible counterfactual. Assign every cost, benefit and recovery once. If displaced work is already missing from the completed-job total, subtracting it again understates the result. None of these business quantities is measured by app opens. Low monthly activity alone does not tell us whether a job was due, completed elsewhere or lost.

1

Jobs due

Eligible vehicle cohort and work

2

Completed here

Across all Auto2000 channels

3

Gross profit per job

Revenue less recognized direct cost

Compare the with-change result with the counterfactual. Keep operating costs below GP separate; count each item once.

Follow completed jobs across channels, then compare with a credible counterfactual.

Limit: Eligible incidence, actual dealer capture, realized job GP and the counterfactual remain unmeasured for the proposed options.

Evidence: MECH-01; extension_hypotheses/MECE_HYPOTHESIS_COVERAGE.csv and accounting review.

(certain) OBSERVED / DERIVED

Useful tools already work before contact

The desktop walkthrough reached five useful entry journeys: vehicle comparison, finance previews, service-cost guidance, a digital-certificate entry and test-drive scheduling previews. Some paths then ask for a plate or VIN; those downstream steps were not tested.

This changes the starting point for the research. A proposed improvement needs to resolve a specific decision or handoff beyond these existing functions. The observations establish reachable page states, not customer comprehension, completed bookings or incremental sales.

Compare hybrid and petrol	Comparison table displayed
Preview financing	Illustration before name or phone
Browse service estimates	Model and variant cost guidance
Find a digital-certificate entry	VIN validation step
Preview a test drive	Branch, date and time windows

The new idea needs a specific improvement beyond the existing route.

Limit: Desktop only. Reaching a page is not a completed customer task or a causal business result.

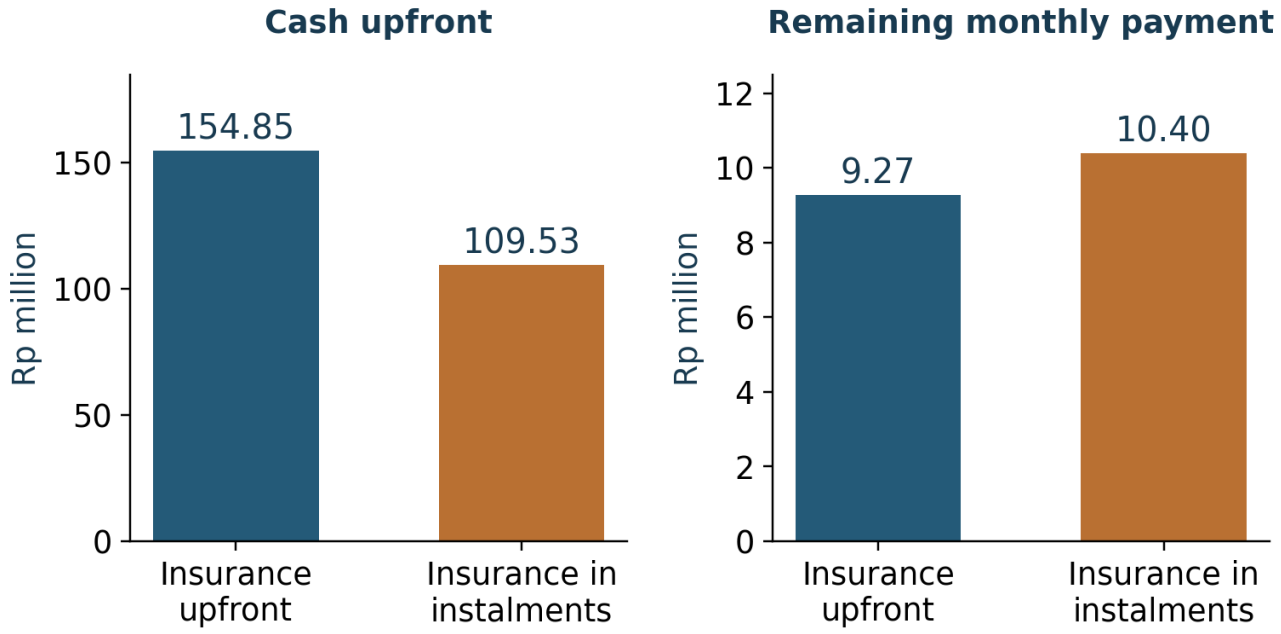
Logged-out desktop observations, 11 November 2026. Evidence UX02-UX05, JX02-JX03 and XT02/XT04/XT06. Safe endpoints only.

(certain) OBSERVED / DERIVED

Insurance timing moves cash between now and later

Changing only insurance-payment timing reduced the displayed cash needed upfront by Rp45.31 million, while raising the remaining monthly instalment by Rp1.13 million. The car, city, base down payment and first-instalment timing stayed the same.

The existing calculator therefore already answers a useful budget question. A further explanation could help a buyer understand cash at delivery and later payments. Whether that explanation improves a purchase decision remains untested. These are conditional simulator outputs, not binding lender or insurance quotes.



Explain cash at delivery and later obligations without inventing an approved rate.

Limit: The source discloses payment timing. No completed finance application, approved rate or customer effect was measured.

UX03-UX04. Innova Zenix G HV CVT; Jakarta Pusat; Rp475.4m; DP 20%; ADDM; 60-month illustration, 59 remaining payments.

(certain) OBSERVED / DERIVED

A BEV page shows a clear label mismatch

The selected bZ4X page labels its fuel system as BEV, yet places 7311 under a tank-capacity field measured in litres. The value was visible in both the page and its accessibility tree.

This gives us a precise information-quality issue to investigate. It does not establish the correct engineering value, the underlying cause or whether a customer changed a decision. The useful question is whether specific labels help a buyer understand the relevant vehicle tradeoff.

Fuel system: BEV

Observed translated label

Displayed value unchanged

7311

Tank capacity (litres)

Literal page value; engineering meaning unverified

Research redraw of two observed fields, not a product screenshot.

Investigate this exact information issue; do not generalize it into a site-wide failure.

Limit: This is one specific page state. No universal website failure or causal customer effect is inferred.

XT05/XTS02; auto2000.co.id/mobil-baru-toyota/p/bz4x/spesifikasi;11 November 2026.

(certain) OBSERVED / DERIVED

Keep preference and purchase as separate outcomes

Cost and fit information can influence what people say they would choose. The ETH working paper reports a shift in BEV purchase preference after personalized information. The Irish study also finds a stated-choice response, while its showroom trial detects no overall change in drivetrain choice among cars sold. That result does not establish a universally zero effect. NIST's battery valuation document is a survey instrument, not a completed premium estimate. These sources leave Auto2000's completed sales and profit effect unknown.

EXT-H-S02: ETH

Stated BEV purchase preference. Reported preference increase; no verified delivery endpoint

EXT-H-S01: Ireland, online

Hypothetical drivetrain choice. Longer-horizon monetary framing affected stated choice

EXT-H-S01: Ireland, showroom

Drivetrain choice among cars sold. No overall effect detected; not proof of zero

EXT-H-S03: NIST DCI 008

Research design. No completed certificate-premium estimate in this document

Better understanding is plausible. A delivery uplift still needs evidence.

Limit: Timing, price, liquidity, available offers, household decisions and fulfillment remain competing explanations; trial precision was not independently re-audited.

Petrov et al.(2025),doi: 10.1016/j.trd.2025.104921; Bernardic et al.(2024); NIST DCI008. Exact locators in extension_hypotheses.

(certain) SELECTED PUBLIC RECORDS

A good rating may be about something else

Sixteen of the 20 displayed Digiroom reviews in the Indonesian Google Play view had four or five stars. Seven describe a specific app interaction. Five do not give enough detail to tell. Four describe another experience or come before a service was used. Even a specific app interaction can be a failed attempt. These records show why favorable stars alone cannot tell us whether someone booked successfully, completed a service or bought a vehicle. They describe this selected view, rather than all users.



7

Specific app interaction

May include a failed attempt

5

Not enough detail

Task or outcome remains unclear

4

Other / not used yet

Car, branch or prospective view

Measure the task and its outcome before treating a rating as evidence of success.

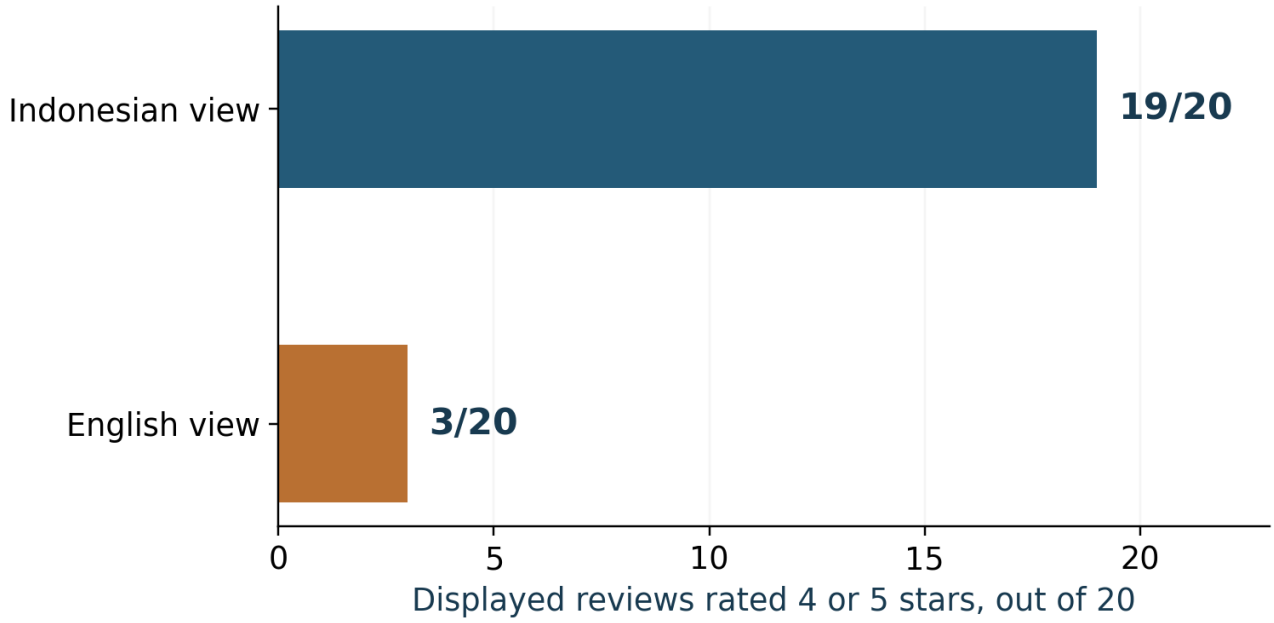
Limit: Selected displayed records, not a representative user sample. AI source coding; no human validation or independently verified customer outcomes.

Specific app interaction includes failures. No completed transaction was independently verified. Evidence: extension_reviews and frozen reviews tables.

(certain) SELECTED PUBLIC RECORDS

Two listing views tell very different stories

The same BYD app has 19 favorable ratings among the 20 records displayed in its Indonesian-language view, but three among the 20 in its English-language view. Both views used the same store country setting and the same four-or-five-star rule. The two sets share no review IDs. Their dates also differ. This contrast shows how the selected view can change the apparent story. It does not identify reviewer nationality or establish a language effect on satisfaction, and it cannot support a brand league table.



Keep source selection consistent before comparing results.

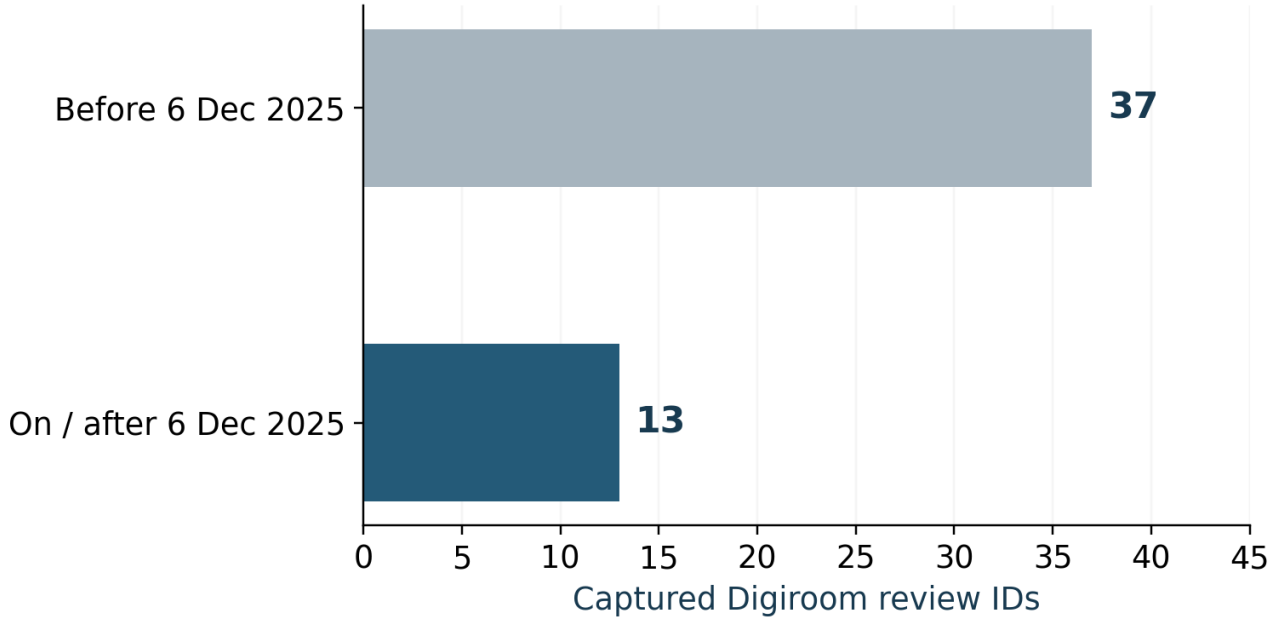
Limit: Both selected cohorts contain 20 records; the favorable threshold is aligned at rating >= 4. Displayed selection and date mix can change; this is not a causal language test.

Selected listing records. Requested language does not establish reviewer location. Evidence: extension_reviews and frozen reviews tables.

(certain) SELECTED PUBLIC RECORDS

Most captured Digiroom reviews predate the relaunch

The first-wave Digiroom collection contains 50 reviews across Google Play and Apple. Thirty-seven carry dates before the 6 December 2025 relaunch; 13 are dated on or after it. Six of those 13 describe a specific app interaction. The rest do not establish one. Older accounts can reveal useful tasks and earlier friction, but they cannot describe the relaunched experience by themselves. A review date is also different from the date of the experience or proof of the installed app build. No before-and-after effect is identified.



Check the current journey directly before using old complaints to set priorities.

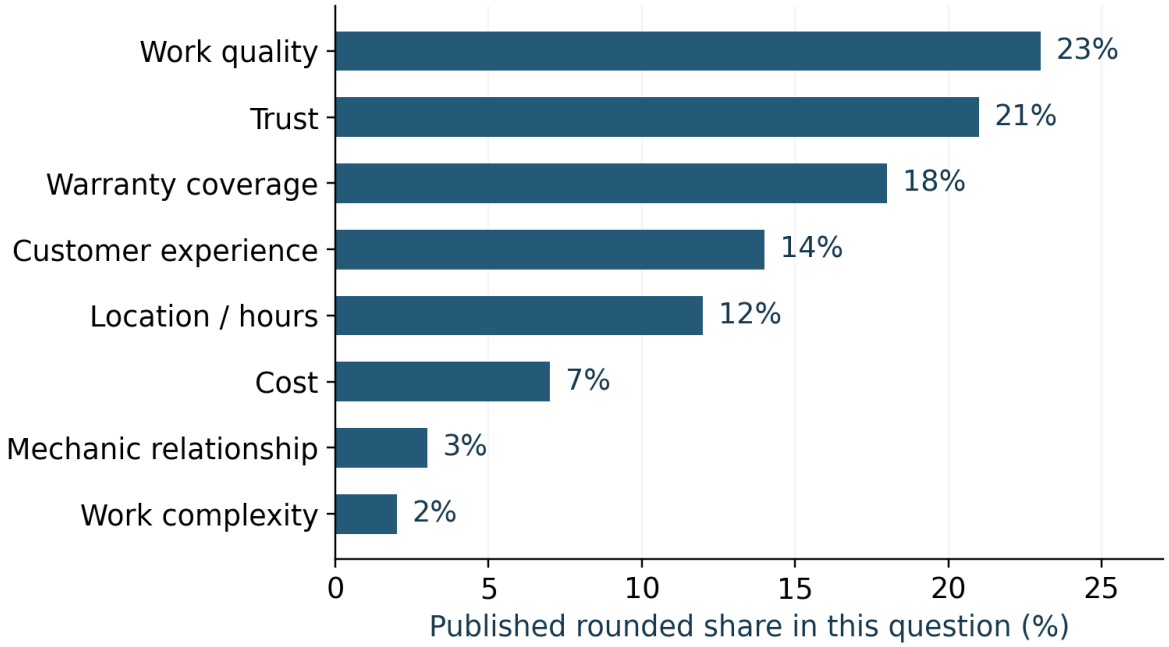
Limit: The date is source-reported review date, not verified experience date or installed-build exposure. Specific interaction includes failures; completion remains unverified.

Source review dates; no installed-build exposure or before/after effect is established. Evidence: extension_reviews and frozen reviews tables.

(certain) PUBLISHED SURVEY

Service confidence extends beyond a booking slot.

In Deloitte's Indonesia service-provider choice question, quality, trust and warranty coverage lead the published reasons. Convenience still matters, but this question does not put it first. These are stated reasons in a national study, not measurements of Auto2000 customers or evidence that one proposed feature changes a sale.



A scheduling idea should explain how it helps the underlying service decision.

Limit: Different survey questions have different bases. A rounded zero in another question does not mean customers never want online booking.

Deloitte 2026 SEA Automotive Consumer Study,p33,Q17,Indonesia n=618. Published rounded shares; fieldworkOct-Nov 2025.

(certain) OBSERVED / DERIVED

Preview works; confirmation remains a later step

The bZ4X journey exposed Juanda's date and time previews before contact. Its FAQ says staff later confirms the schedule and unit availability. That confirmation was not observed because no request was submitted.

Branch context also survived two tested handoffs: Juanda with bZ4X, then Cempaka Putih with Hiace. The second route changed the preceding branch selection. These counterexamples shift the question toward actual availability and fulfilment. They do not establish a universal handoff problem, a confirmed vehicle or a completed visit.

- **Preview branch, date and time**
  Observed on desktop
- **Submit request**
  Not performed
- **Staff confirms unit and schedule**
  Stated in FAQ; not observed
- **Attend and complete the job**
  Not observed

A tentative window and a confirmed unit are different promises.

Limit: Desktop only. Zero mobile-emulated and native-app journeys: local browser launch blocked by socket permissions; native DevTools blocked by organization policy; no available native emulator identified.

Logged-out desktop, 11 November 2026; XT06 and XT08. Juanda date/time preview and booking FAQ; two actual branch links checked after loading.

(certain) OBSERVED / DERIVED

Branch and vehicle context survived two tested links.

Two actual branch links carried their selections into the test-drive route. Juanda opened with bZ4X; Cempaka Putih opened with Hiace and changed the preceding branch selection. These are useful counterexamples to a blanket claim that handoffs always lose context. They do not establish technician competence, current unit stock, a confirmed slot or successful fulfilment.

Actual branch link		Loaded selection
Juanda / bZ4X	→	Juanda / bZ4X
Cempaka Putih / Hiace	→	Cempaka Putih / Hiace

Keep the functioning baseline and look for the narrower unresolved handoff.

Limit: Two logged-out desktop routes; no request or personal data submitted. Mobile and native-app behavior remain untested.

XT08, XTS10/XTS12. Branch codes T151 and T155. Detailed URLs and observations in extension_journeys.

(certain) PUBLISHED / DERIVED

Longer coverage does not always mean more covered visits.

Current Toyota material gives T-CARE seven scheduled labor-and-parts events within three years or 60,000 km. T-CARE Lite covers labor for seven events; its parts package is separate. Legacy GBSB allows six events within four years or 50,000 km. Its published calendar lists events through month 30. Headline years therefore do not identify covered visit count, the customer's actual payment date or dealer gross profit. Eligibility and specified work still govern.

T-CARE

3 years / 60,000 km maximum

1 2 3 4 5 6 7

Labor + specified parts

T-CARE Lite

3 years / 60,000 km maximum

1 2 3 4 5 6 7

Labor only; base programme

Legacy GBSB

4 years / 50,000 km maximum

1 2 3 4 5 6

Labor + specified parts

Keep programme eligibility, covered work and event count separate from headline years.

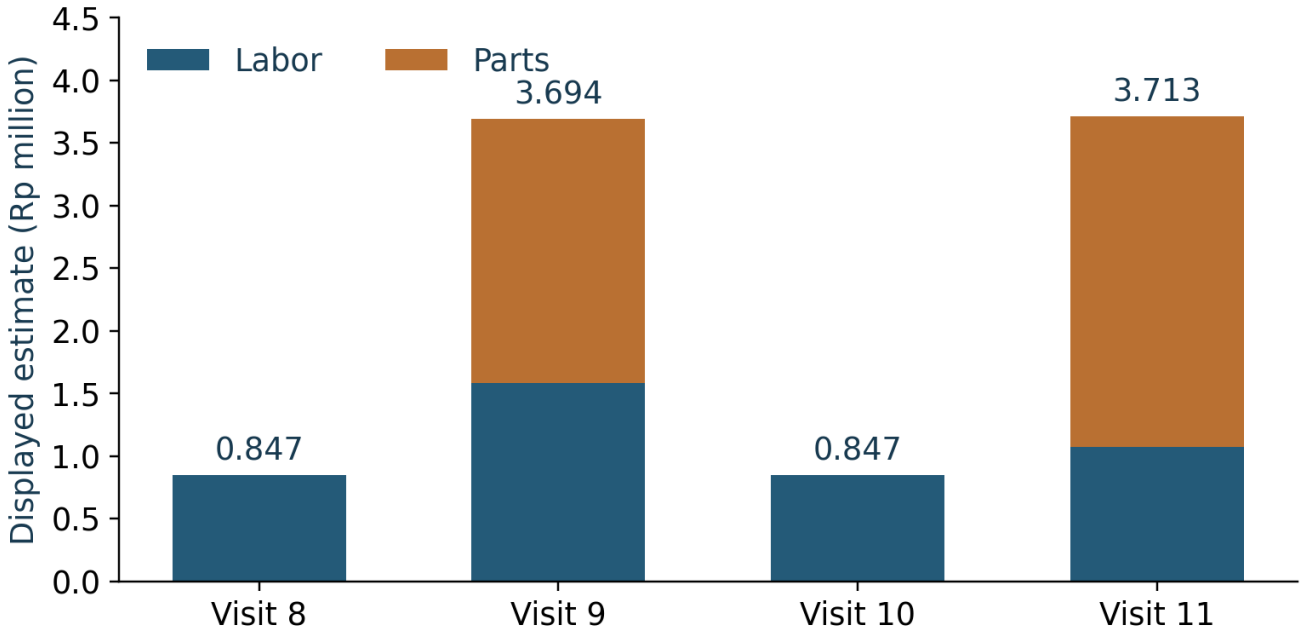
Limit: Labour coverage is not universal parts coverage. Maximum duration is not an observed return date.

Source: Toyota T-CARE programme page;2026XTRAannouncement; legacy GBSB PDF p1. Time/km means whichever first; eligibility, specified work and model terms apply.

(certain) PUBLISHED / DERIVED

Later service bills vary within one model.

The public Toyota tool lists bZ4X Panoramic View estimates from Rp847,000 at visits 8 and 10 to Rp3,713,000 at visit 11. Labor and parts sum to Rp9,101,000 across the four events. The opened note dates these estimates to March 2024 and allows dealer and additional-work differences. These are useful illustrations of uneven bills, not current invoices, total ownership cost, dealer margin or a forecast of customer spending.



A dated retail estimate can inform a conversation; it cannot reveal dealer gross profit.

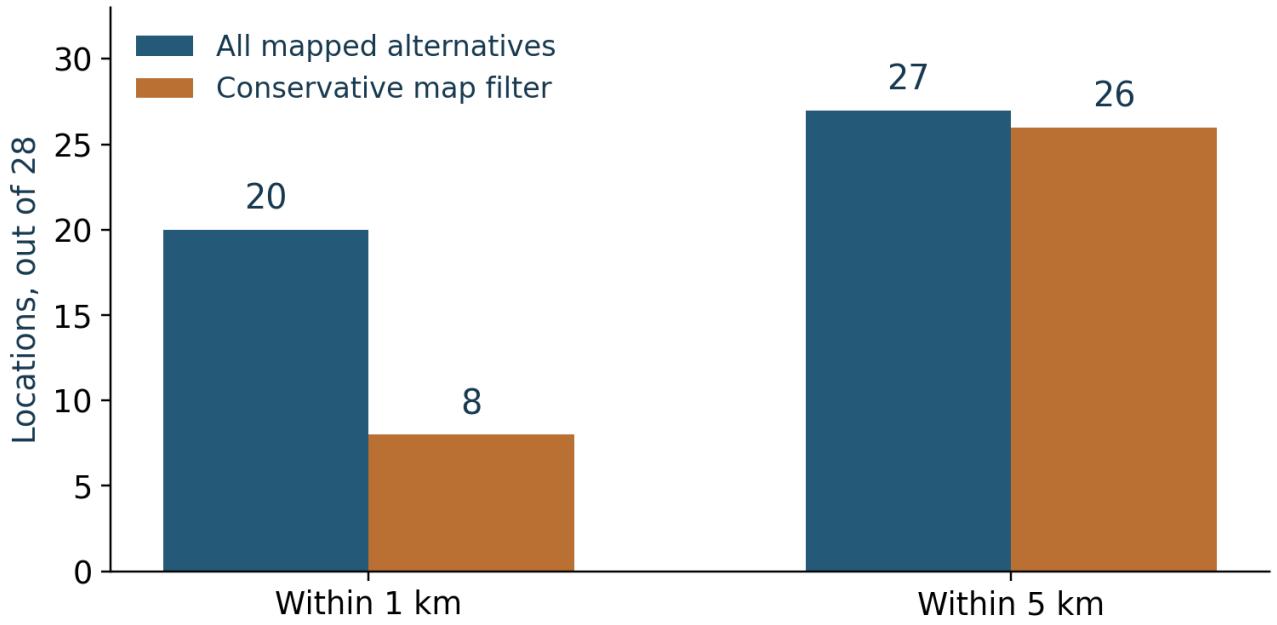
Limit: Do not display these as 2026 prices. Do not multiply by case line margins to create EV-specific GP.

Source: Toyota model/variant cost tool, XT04, observed 11Nov2026. Estimates explicitly based on March 2024; tax included; dealer/extra-work caveats. Not a current quote or GP.

(certain) DERIVED SENSITIVITY

Nearby on a map does not establish usable access.

Across 28 Auto2000-name locations, a conservative map filter cuts nearby options within 1 km from 20 to 8. At 5 km, the count changes only from 27 to 26. The nearest-option median rises from 0.323 to 1.413 km. This shows that close-range convenience depends on which sites we count. The filter still does not verify whether a driver can use them.



The 5 km result holds up. The 1 km result changes sharply.

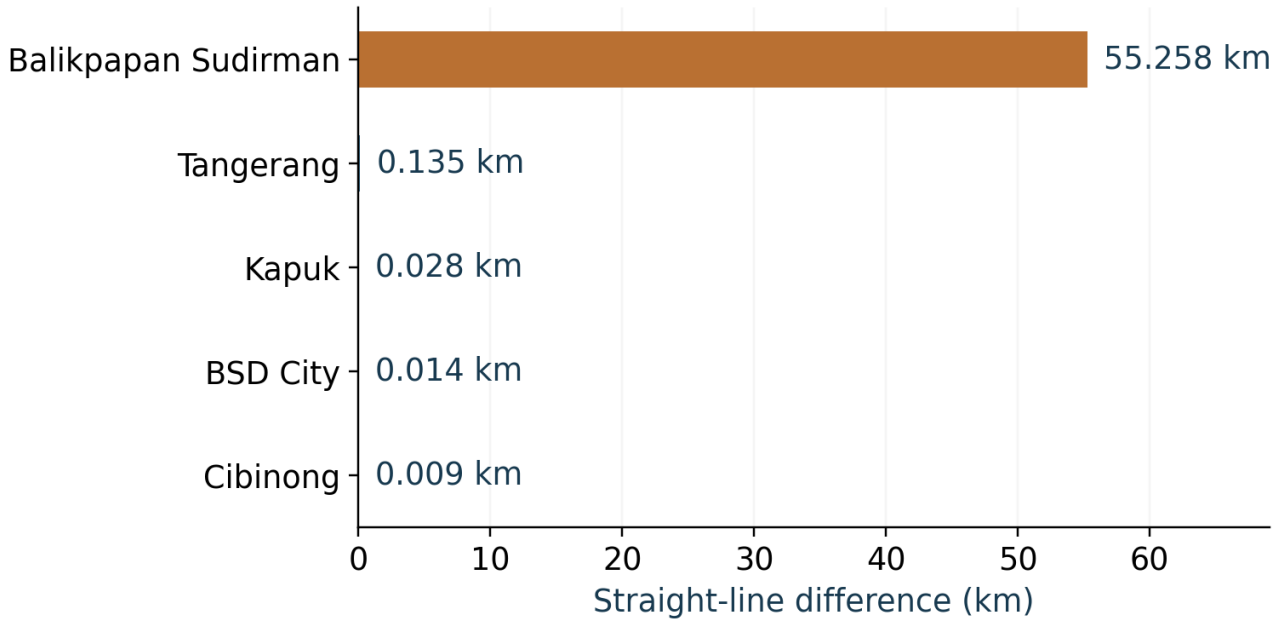
Limit: The filter uses operator and venue names. It does not verify access, connector compatibility, opening hours or uptime.

Original source coordinates. All distances are straight-line; 28 name matches are not the full Auto2000 branch network. Source: SPKLU June 2026; Wiputra and Santiadi, DOI10.17632/z2bvd5r574.2, CC BY-NC 3.0. Original derived analysis.

(certain) DERIVED SENSITIVITY

One published location differs by 55 kilometres.

We compared five charging-source points with current official branch directions. Four pairs are within 135 metres. Balikpapan Sudirman differs by 55.26 km. Replacing that point changes the nearest-site result, so a broad map summary can hide a local problem. This comparison identifies a published-data conflict. It does not tell us where the charger was physically located on either date.



Check the location before trusting a proximity score.

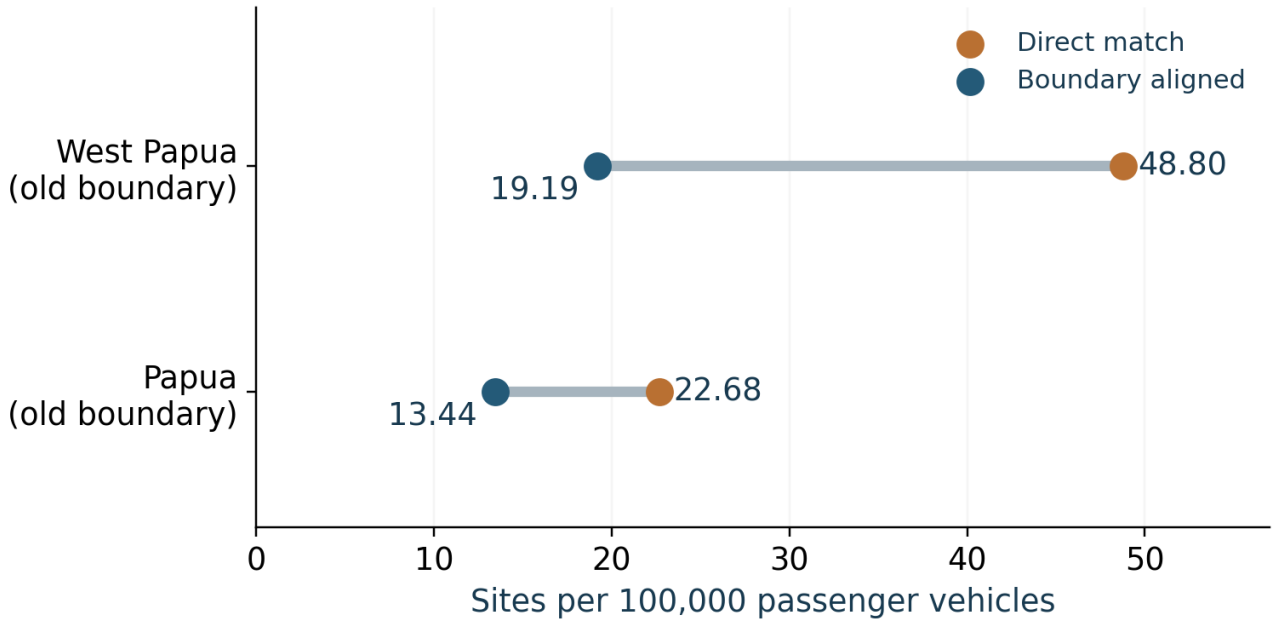
Limit: Official branch directions are a second published source, not field confirmation of charging hardware.

Charging-source snapshot: 16 June 2026. Official branch pages checked: 11 November 2026. Preserve both coordinate versions. Source: SPKLU CC BY-NC 3.0; current official Auto2000 branch directions. Original comparison.

(certain) DERIVED SENSITIVITY

Boundary matching changes the provincial ranking.

The charging source uses older provincial boundaries. The 2025 vehicle table separates four newer Papua provinces. A direct code match drops 70,800 cars from the denominator. Pooling those provinces back into the older boundaries lowers West Papua's ratio from 48.80 to 19.19 and Papua's from 22.68 to 13.44. The original apparent ranking depended on a mismatched join.



West Papua falls from rank 1 to rank 10 after boundary alignment.

Limit: These ratios use all passenger vehicles, not BEVs; boundary alignment does not measure charging adequacy.

June 2026 mapped sites divided by 2025 BPS/Polri stocks. This corrects the first-wave direct-join interpretation.
Source: SPKLU CC BY-NC 3.0; BPS/Polri published vehicle stocks. Original boundary-alignment analysis.

(certain) DERIVED SENSITIVITY

Four charging counts mean four different things

The same source contains 2,454 station records, 3,767 charger-box records, 7,593 reconstructed charger units and 4,665 connectors. These quantities describe different levels of the source data. They cannot be added or substituted for one another. None tells us how many compatible vehicles can charge at once today. Choosing a different unit can also change provincial rankings.

2,454

Station records

Consolidated source locations

3,767

Charger-box records

Separate hardware records

7,593

Source charger units

Counts reconstructed by the source

4,665

Source connectors

Recorded connector counts

A site count is not a capacity measure.

Limit: Hardware counts do not establish usable power, simultaneous charging capacity, vehicle compatibility or uptime.

Source snapshot: 16 June 2026. Exact coordinate duplicates were already consolidated by the source. Source: Wiputra and Santiadi, Mendeley V2, DOI10.17632/z2bvd5r574.2, CC BY-NC 3.0.

(certain) OBSERVED / DERIVED

Toyota retention and Auto2000 capture can differ.

Toyota already offers fleet telematics and an economical parts/repair ecosystem. These are baselines for new reminder or older-vehicle ideas. A customer who stays with Toyota through another authorised or economical channel may support the ecosystem without producing the same Auto2000 branch ticket. The customer, decision owner, payer and business perimeter need to stay visible.

Customer outcome

A suitable, affordable job is completed for the person or fleet.

Toyota ecosystem

May retain the owner through different brands, partners and economical repair channels.

Auto2000 business

Needs attributable completed work and contribution at its own stated perimeter.

Define whose outcome improves before calculating an incremental benefit.

Limit: Public programme descriptions do not establish local adoption, referral fees or dealer profit. Europe-only warranty terms must not be transferred to Indonesia.

Extension stakeholder audit: G-Fleet, T-OPT/OtoXpert, partner roles and geographic warranty checks. 18 targeted URLs; 21 scoped observations.

(certain) OBSERVED BASELINE / UNTESTED MECHANISM

Certificate value depends on its use in a real decision

Toyota already provides a digital Hybrid Health Check certificate route. The remaining question is whether trustworthy evidence reaches a buyer or appraiser and changes a real decision. Keep five links separate: valid measurement, understandable result, credible source, acceptance by the decision owner, and a changed transaction. Our Audi telemetry analysis tests data-handling limits, not Toyota certificate accuracy. Its confidential capacity reference and asynchronous signals prevent a validated health score here. NIST's usefulness survey does not fill the missing appraisal or price link.

Measure	Technically valid result
Understand	Scope, age and limits
Trust	Credible issuer
Accept	Appraiser uses it
Change outcome	A transaction changes

A retrieval route is a baseline. Accepted evidence and a changed transaction are later links.

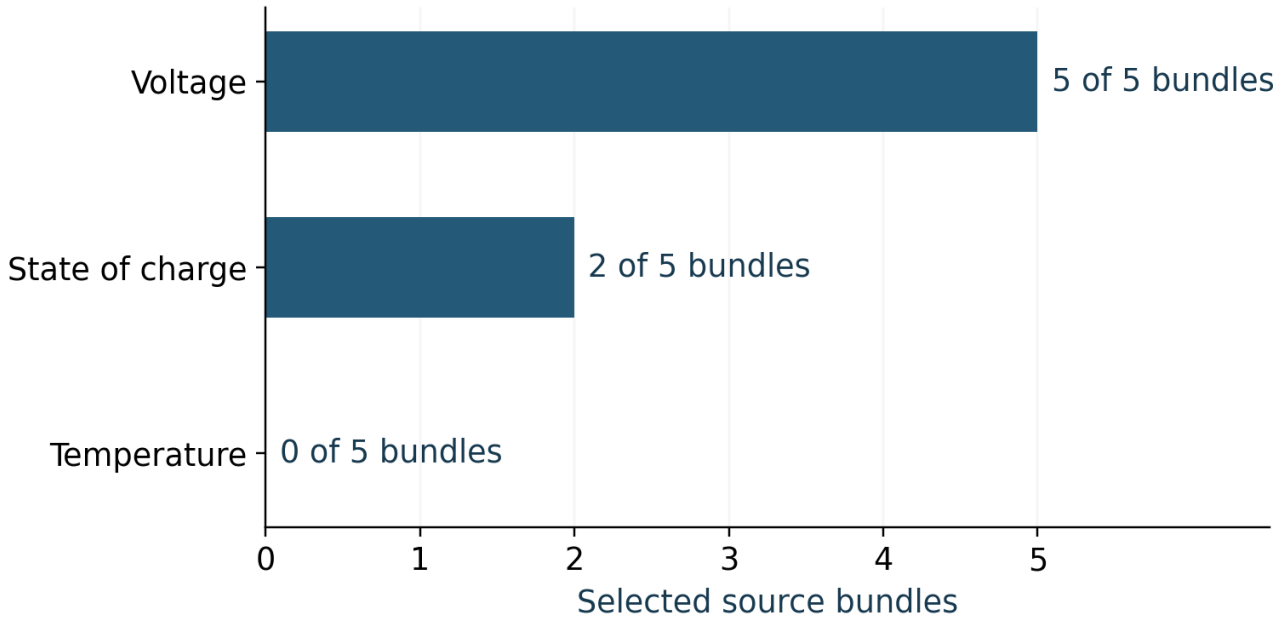
Limit: The open question is valid, understandable, accepted evidence at the decision. A price premium is only one possible outcome and remains unmeasured.

JX02; NIST TN2328 and DCI008; extension_hypotheses and extension_battery. No certificate downloaded or appraisal observed.

(certain) AUXILIARY MEASUREMENT TEST

The large telemetry subset still lacks a public health reference.

One historical Audi in California supplies 56.5 million stored electrical and BMS signal values. They support signal analysis, not a Toyota health certificate. The paper's capacity reference is withheld. Current and voltage share a clock in all five bundles, but SOC does so in only two and temperature in none. The measurements need careful alignment before any health claim can be tested.



The missing reference matters more than the row count.

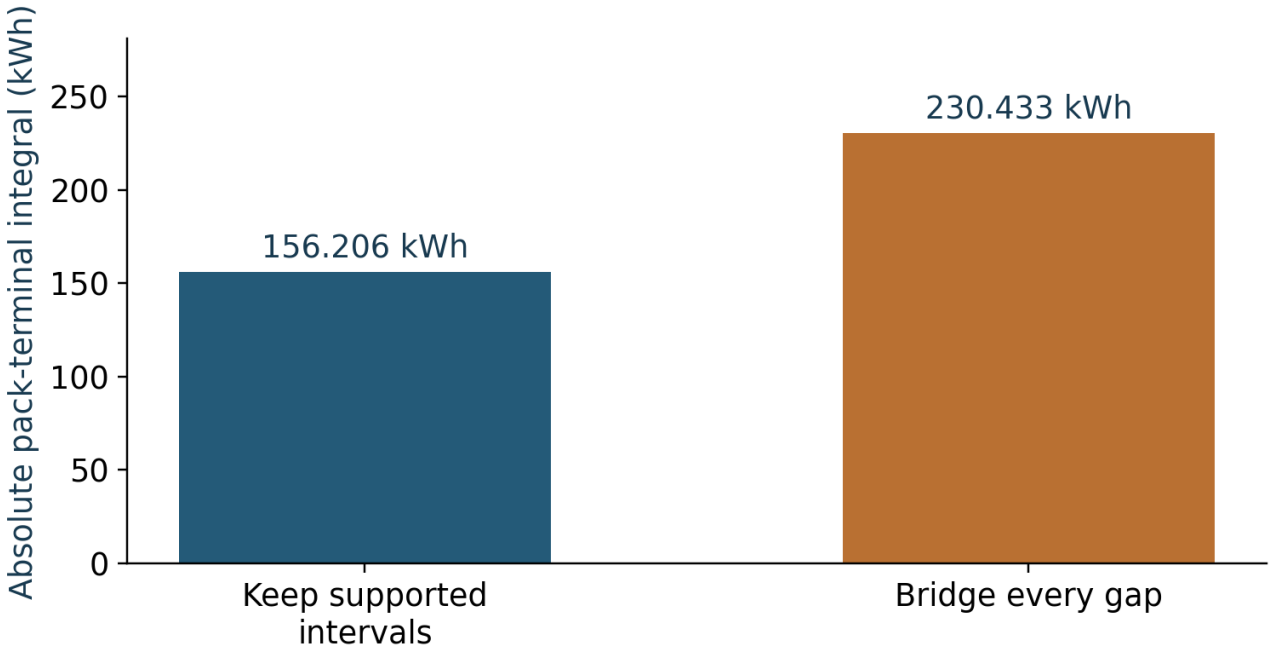
Limit: Counts exclude clocks and epoch metadata. Values are not independent customers, trips or synchronized rows.

Five folder windows from a single Audi e-tron, 2019-2020. SOC means state of charge, not state of health. Source: Allam et al.(2023), Mendeley V2, doi: 10.17632/7vdkzpnjgj.2, CC BY-NC3.0.

(certain) AUXILIARY MEASUREMENT TEST

Bridging recording gaps inflated one result by 47.5%.

Across five bundles, the one-second gap rule retains 51.97 supported hours. Folder16 alone spans 50.82 calendar days, yet contains only 17.19 such hours. Bridging every gap inflates Folder2’s absolute terminal integral from 156.206 to 230.433 kWh. The extra 47.52% comes from an unsupported calculation across missing intervals. The tested 1-, 30- and 300-second limits agree here.



Do not treat days between logs as measured driving.

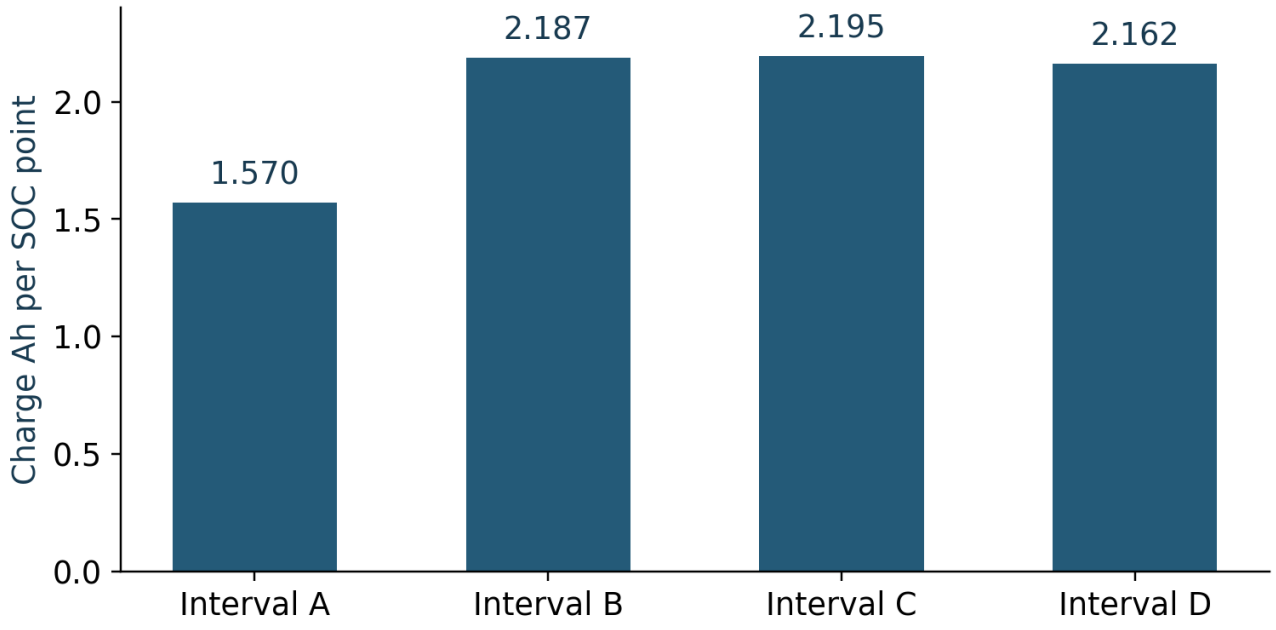
Limit: This is a signed-signal throughput calculation, not battery capacity, charger wall energy, efficiency or a production-system failure.

Absolute integral adds positive and negative energy magnitudes. Current and voltage clocks match in these files.
Source: Allam et al.(2023), Mendeley V2, doi: 10.17632/7vdkzpnjgj.2, CC BY-NC3.0.

(certain) AUXILIARY MEASUREMENT TEST

Four partial-charge slopes are not four health scores.

Four supported charge intervals give 1.570 to 2.195 Ah per reported SOC percentage point. Their current, temperature and SOC windows differ. Every stored current value lies on an integer-ampere grid, but the sensor's accuracy bound is unknown. These ratios describe the selected intervals. Rescaling them into battery capacity or state of health would add an assumption this dataset cannot validate.



A partial charge cannot become a health score without validation.

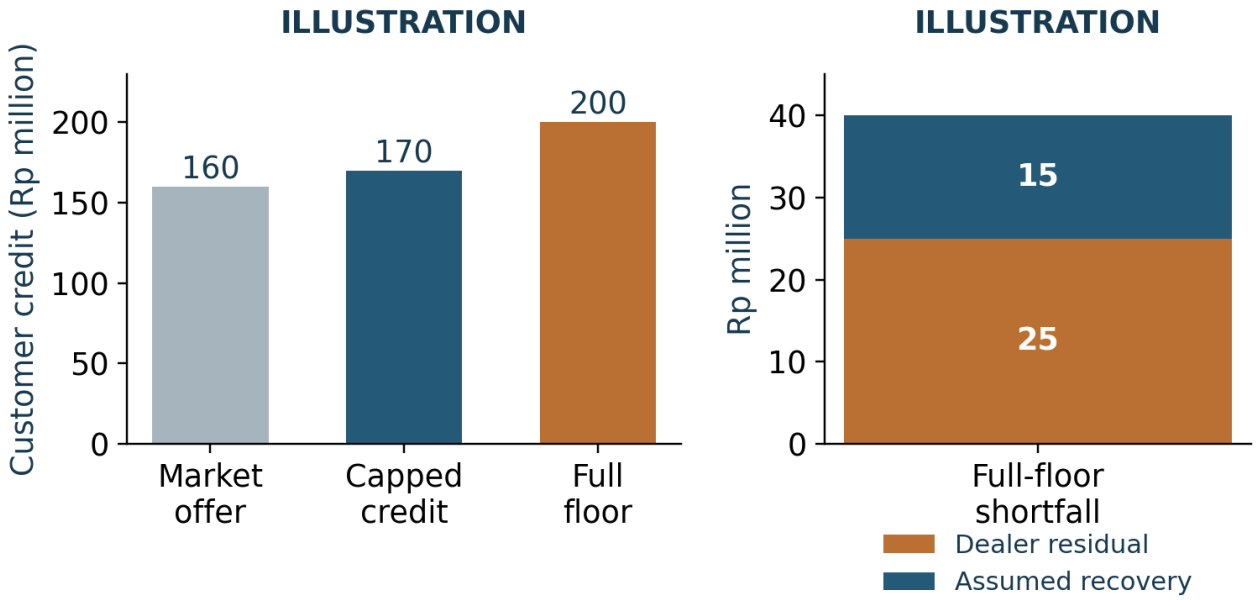
Limit: Four nonrandom intervals from one vehicle cannot establish degradation, capacity, health-score accuracy or a temperature effect.

SOC is state of charge. Common observed support only; no endpoint extrapolation. The health reference is not public. Source: Allam et al.(2023), Mendeley V2, doi: 10.17632/7vdkzpnjgj.2, CC BY-NC3.0.

(certain, conditional) ILLUSTRATION ONLY

A capped top-up does not create a full price floor.

In this constructed example, the capped top-up adds Rp10 million to a Rp160 million offer, producing Rp170 million of customer credit. A separate full-floor promise requires Rp200 million. Its Rp40 million shortfall leaves Rp25 million with the dealer after the assumed Rp15 million insurer layer, using a Rp10 million retention. That arithmetic excludes contribution and running costs. No actual policy, underwriting appetite or approved Auto2000 offer is established.



State the customer promise and retained exposure before discussing who might fund it.

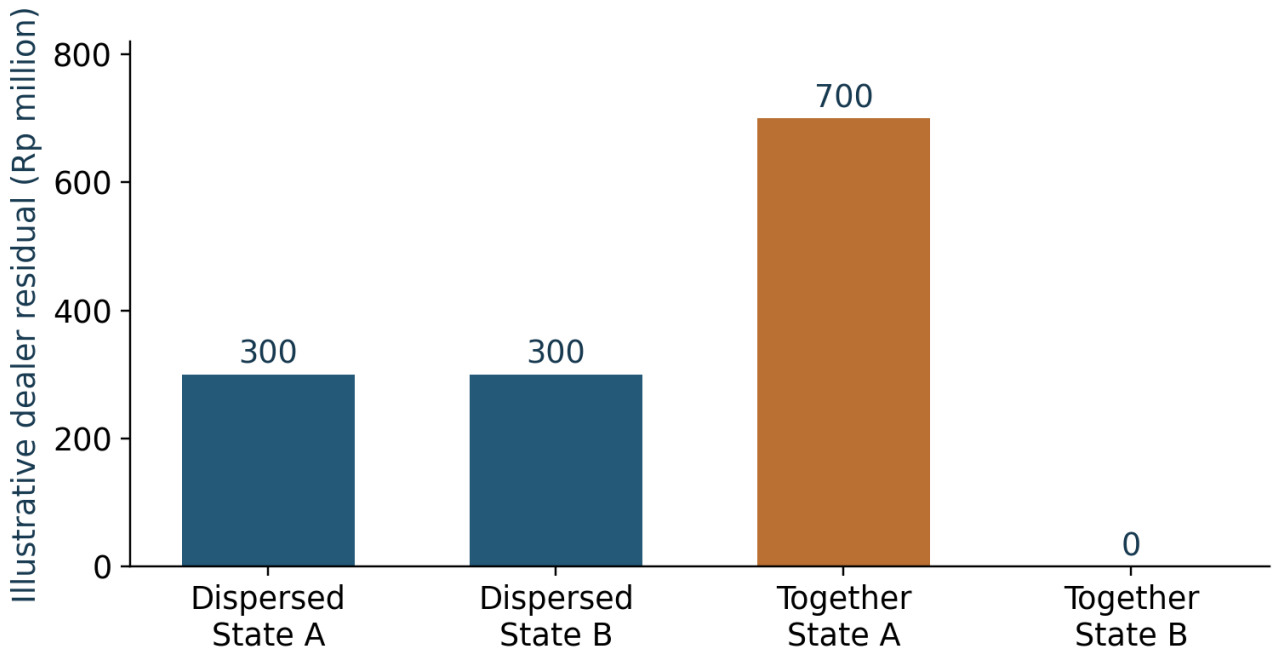
Limit: Do not call the capped top-up a guaranteed Rp200m floor. The insurer layer is assumed, conditional and not a quote.

Constructed illustration using declared floor/cap/retention/layer assumptions. All losses assumed eligible and aggregate cover available; premium, expenses and timing excluded. No available policy verified.

(certain, conditional) ILLUSTRATION ONLY

The same offers can produce a larger peak loss.

These constructed worlds use the same twenty units and the same pooled low and high offers across two stress states. Dispersion puts ten low offers in each state; synchronization puts all twenty in one. Under the assumed Rp100 million aggregate recovery limit, the highest dealer residual rises from Rp300 million to Rp700 million. No probability is assigned. This exposes a limit of per-unit reasoning without estimating actual correlation, expected loss or insurance availability.



A limit on insurer payment does not cap a separate full-floor promise.

Limit: Four states are not four observed markets or four probabilities. Eighty constructed rows do not describe eighty customers.

Constructed stress arrangement; same pooled unit-state offers, 20 units/state. Recovery is conditional and capped at assumed Rp100m/state. No probabilities, quote, actual customers or expected-loss estimate.

(certain) MECE COVERAGE AUDIT

MECE reveals lifecycle gaps as well as more features.

The lifecycle audit organizes 25 customer or fleet outcomes plus one institutional funding layer across ten grouped decision-owner roles. It also highlights recall completion, delivery closure, reasoned redress and terminal ownership or account exit. These are coverage questions, not proof that Auto2000 lacks the relevant functions. Safety-campaign completion should not be treated as discretionary paid upsell.

Choose and acquire

Delivery closure is a separate job

Maintain and repair

Model, programme and payer matter

Protect and resolve

Recall completion and reasoned redress

Replace or exit

Appraisal, custody and account exit

Use the matrix to find an unobserved job and its decision owner, then test the gap.

Limit: The 260 matrix cells are not 260 independent needs. Recall is one separate research candidate; no solution is selected.

Extension stakeholder audit: 26 analytical job units, 10 grouped roles, 18 finer stakeholder lenses. EXT-OPT01 builds on existing VIN lookup and booking.

(guessing) POSSIBLE DIRECTIONS

Vehicle and finance choices

Possible directions for your choice. Existing baseline, possible addition and a reason it might fail are kept together.

SOL-01 | A use-based vehicle decision brief

Already: Auto2000 already compares vehicles and finance costs; Toyota publishes service estimates with an older date.
Possible: Help a household compare the actual vehicle, budget, journeys and charging arrangement using clearly dated assumptions.
Could fail if: Price or stock decides the purchase, or existing advice already settles the choice.

SOL-02 | Practical home-charging readiness

Already: PLN and qualified installers already support home charging; a nearby charger does not establish permission to use it.
Possible: Clarify parking rights, building permission, electrical capacity and vehicle compatibility for the customer's actual charging arrangement.
Could fail if: Existing providers resolve readiness, public charging already fits, or the required permission is impossible.

SOL-03 | Finance status and total obligations

Already: The finance simulator works before contact entry; SEVA already offers routes toward a lender's approval decision.
Possible: Explain which amounts are estimates, what remains unapproved and the customer's full obligations under an actual offer.
Could fail if: The customer already understands the offer, or affordability and underwriting determine the outcome.

SOL-04 | Purchase versus subscription on full obligations

Already: Subscription and commercial rental programmes exist, with different included costs, usage rules and ownership obligations.
Possible: Compare eligible purchase and subscription contracts for the same use, including end-of-term costs and restrictions.
Could fail if: No comparable eligible contract exists, or existing quotes already make the choice clear.

All benefits remain hypotheses. Full evidence, dependencies and alternative explanations: reports/SOLUTION_MENU.json and workstreams/extension_stakeholders/all_27_option_audit.json.

(guessing) POSSIBLE DIRECTIONS

Service work, access and parts

Possible directions for your choice. Existing baseline, possible addition and a reason it might fail are kept together.

SOL-05 | Understandable service work and bill changes

Already: Work quotations and service records may already exist; this research did not inspect private repair orders.
Possible: Make necessary work, provisional estimates, approved changes and the final bill understandable within the same service case.
Could fail if: Current explanations are sufficient, or the dispute concerns actual repair quality rather than unclear information.

SOL-06 | An appointment commitment customers can interpret

Already: Booking and guest routes exist; a requested date can still require branch review and confirmation.
Possible: Make clear whether the customer has requested a visit, received acceptance or secured a specific service commitment.
Could fail if: Current confirmations already work, or missing technicians, parts and physical capacity remain the binding problem.

SOL-07 | Visit preparation and conditional timing

Already: Current forms and partner instructions already request vehicle details, documents and other task-specific prerequisites.
Possible: Explain what must happen before a visit, inspection or advertised completion time becomes a realistic commitment.
Could fail if: Preparation failures are uncommon, existing advisers resolve them, or extra instructions only increase effort.

All benefits remain hypotheses. Full evidence, dependencies and alternative explanations: reports/SOLUTION_MENU.json and workstreams/extension_stakeholders/all_27_option_audit.json.

(guessing) POSSIBLE DIRECTIONS

Service work, access and parts (continued)

Possible directions for your choice. Existing baseline, possible addition and a reason it might fail are kept together.

SOL-08 | A service route matched to the vehicle and job

Already: Branch filters, guest booking, home service and body repair routes exist; economical workshops add another existing choice.
Possible: Clarify which available provider can handle this model and job, including specialist or lower-cost care when relevant.
Could fail if: Current routing already works, or reliable capability and availability information cannot be obtained.

SOL-09 | Compatible parts with clear coverage consequences

Already: Official parts channels and economical T-OPT parts exist; model lists do not establish every variant's fit or stock.
Possible: Explain the actual part's compatibility, availability, price and any protection consequences before the customer approves work.
Could fail if: Existing parts advice suffices, or price and unavailable supply matter more than information.

All benefits remain hypotheses. Full evidence, dependencies and alternative explanations: reports/SOLUTION_MENU.json and workstreams/extension_stakeholders/all_27_option_audit.json.

(guessing) POSSIBLE DIRECTIONS

Ownership records and protection

Possible directions for your choice. Existing baseline, possible addition and a reason it might fail are kept together.

SOL-10 | A vehicle-specific protection decision

Already: Toyota already offers several care and warranty programmes, each with its own eligibility and payment rules.
Possible: Show which benefits apply to this vehicle now, what expires next and what any paid continuation actually covers.
Could fail if: The customer already understands the benefits, is ineligible, or receives better value from existing alternatives.

SOL-11 | Protection after an ownership or use change

Already: Toyota allows remaining warranty to continue for eligible later owners; other programmes have different transfer and use conditions.
Possible: Explain what a sale, shared use or commercial use changes under the actual contract and ownership record.
Could fail if: Existing ownership updates resolve the change, or the relevant contract imposes no consequential restriction.

SOL-12 | A battery result explained in buyer terms

Already: Toyota offers eligible battery checks and digital certificates; a certificate does not automatically guarantee resale value.
Possible: Explain what a specific test measures, its date and limits, and how warranty or appraisal decisions remain separate.
Could fail if: The measure is unreliable, existing explanations suffice, or the result changes no meaningful decision.

SOL-13 | A coherent ownership record across jobs

Already: mToyota advertises service history; Astra describes customer systems, but private record completeness was not tested.
Possible: Keep permitted vehicle information useful when the customer moves between service, protection and replacement decisions.
Could fail if: Existing records are adequate, information cannot be shared appropriately, or continuity changes no outcome.

All benefits remain hypotheses. Full evidence, dependencies and alternative explanations: reports/SOLUTION_MENU.json and workstreams/extension_stakeholders/all_27_option_audit.json.

(guessing) POSSIBLE DIRECTIONS

Replacement value

Possible directions for your choice. Existing baseline, possible addition and a reason it might fail are kept together.

SOL-14 | Condition evidence accepted in an appraisal

Already: Digital certificates, used-car inspections and BEV trade-in selection exist; appraiser acceptance of each report remains unknown.
Possible: Determine whether a buyer would use trustworthy condition evidence when making a real inspection and purchase offer.
Could fail if: The buyer does not accept the evidence, or market price dominates any useful condition information.

SOL-15 | Trade-in value with explicit uncertainty

Already: Trade-in routes already separate an initial estimate from inspection and a later offer; competing bids remain available.
Possible: Explain why an estimate can change, which conditions matter and what obligations remain at the actual sale.
Could fail if: The process is already understood, or dissatisfaction concerns the market price rather than uncertainty.

SOL-16 | A capped replacement incentive

Already: Appraisals and trade-in allowances exist; no Auto2000 contract for this proposed extra credit was verified.
Possible: Offer a clearly limited replacement credit, with its cap understood as different from a guaranteed future value.
Could fail if: The subsidy mainly rewards customers who would return anyway, or its cost exceeds additional contribution.

SOL-17 | A contractual minimum trade-in value

Already: Some manufacturers offer conditional resale guarantees; this does not establish an equivalent Auto2000 promise or funding.
Possible: Assess a genuine minimum-value commitment, including who owes the full shortfall and which conditions apply.
Could fail if: The guarantor cannot afford or reliably honor the obligation when vehicle prices fall together.

All benefits remain hypotheses. Full evidence, dependencies and alternative explanations: reports/SOLUTION_MENU.json and workstreams/extension_stakeholders/all_27_option_audit.json.

(guessing) POSSIBLE DIRECTIONS

Channel and relationship

Possible directions for your choice. Existing baseline, possible addition and a reason it might fail are kept together.

SOL-18 | One clear route for each ecosystem job

Already: Digiroom, Toyota apps, advisers and partner services already provide overlapping routes for different ownership jobs.
Possible: Help customers reach the correct existing service and decision owner without assuming that every route needs consolidation.
Could fail if: Customers already find the right route, or clearer navigation cannot resolve the underlying constraint.

SOL-19 | Customer context carried into assisted help

Already: Dealer customer systems and insurer claim channels already exist; shared ownership does not prove useful information transfer.
Possible: Carry permitted case details into assisted help, while lenders and insurers retain their own decisions.
Could fail if: Existing case owners already have the context, or consent and partner authority prevent a useful handoff.

SOL-20 | Access that fits household or fleet roles

Already: Toyota already allows another person to deliver a vehicle; G-Fleet provides an existing fleet-monitoring service.
Possible: Clarify who can view, approve and carry out a task without assuming the driver owns the vehicle.
Could fail if: Existing representative processes suffice, or customer and employer permissions cannot support the proposed access.

SOL-21 | Support timed to meaningful ownership jobs

Already: Service reminders and loyalty programmes already exist; low-use PHEVs can still have engine and fuel-care needs.
Possible: Link support to meaningful ownership jobs, including low-use care, without treating frequent app visits as value.
Could fail if: Current reminders already work, extra contact adds no value, or fewer messages would miss necessary care.

All benefits remain hypotheses. Full evidence, dependencies and alternative explanations: reports/SOLUTION_MENU.json and workstreams/extension_stakeholders/all_27_option_audit.json.

(guessing) POSSIBLE DIRECTIONS

Emergency and connected use

Possible directions for your choice. Existing baseline, possible addition and a reason it might fail are kept together.

SOL-22 | Local emergency help with explicit scope

Already: OEM and insurer emergency services already exist, with different locations, hours, incident types and entitlement conditions.
Possible: Clarify who can actually dispatch suitable local help, what it covers and any customer charges.
Could fail if: Existing helplines resolve the task, or reliable recovery supply remains unavailable despite clearer information.

SOL-23 | Mobility continuity during repair

Already: Some programmes already offer conditional transport or replacement vehicles; household backup cars and existing rentals are alternatives.
Possible: Address a specific essential-mobility shortfall during repair, with clear eligibility, available supply and responsibility for costs.
Could fail if: Existing alternatives already suffice, or the required replacement vehicle cannot be supplied economically.

SOL-24 | Understandable vehicle-linked account setup

Already: OEM vehicle-linked services and dealer-assisted setup exist; this research did not authenticate accounts or install native apps.
Possible: Explain the correct vehicle link, ownership checks and permitted support route when setup does not work.
Could fail if: Current setup support succeeds, or the dealer lacks a useful authorized role in resolving the issue.

SOL-25 | An informed connected-service renewal choice

Already: Connected services and renewal routes already exist; fleet and private-vehicle contracts can have different terms.
Possible: Make the actual renewal price, included functions and contract owner clear for this vehicle and customer.
Could fail if: The current renewal route already answers the question, or Auto2000 has no useful authorized role.

All benefits remain hypotheses. Full evidence, dependencies and alternative explanations: reports/SOLUTION_MENU.json and workstreams/extension_stakeholders/all_27_option_audit.json.

(guessing) POSSIBLE DIRECTIONS

Insurance and risk transfer

Possible directions for your choice. Existing baseline, possible addition and a reason it might fail are kept together.

SOL-26 | A voluntary EV protection choice with real terms

Already: EV-specific insurance and claims channels exist; battery wear, manufacturing defects and resale losses are different risks.
Possible: Help customers compare voluntary cover using the actual vehicle, insured events, exclusions, deductibles and repair-network conditions.
Could fail if: Existing policy explanations suffice, the relevant risk is excluded, or no valid distribution arrangement exists.

SOL-27 | Investigate B2B residual-risk transfer

Already: Some international insurers cover agreed future-value risks for businesses; an Indonesian syariah contract was not confirmed.
Possible: Assess whether an actual insurer would accept a defined part of the dealer's promised future-value exposure.
Could fail if: No suitable local contract exists, premiums overwhelm the benefit, or uncovered obligations remain unaffordable.

All benefits remain hypotheses. Full evidence, dependencies and alternative explanations: reports/SOLUTION_MENU.json and workstreams/extension_stakeholders/all_27_option_audit.json.

(certain) SELECTED PUBLIC RECORDS

Different evidence answers different questions

The direct review collection contains 358 unique platform review IDs: 210 from the first wave and 148 added later. Separately, the narrative work contains 95 annotated units drawn from 38 public URLs. A unit can be a personal account, advice or a research finding. These quantities describe different objects and should not be added into one customer count. Several passages can come from the same document, and source IDs do not verify unique people. The package supports diagnosis and further questions rather than verified customer outcomes.

358

Unique app-review IDs

210 first-wave +148 additional; selected records, not verified people.

95

Narrative annotations

Across 38 source URLs; not 95 independent customers.

Keep these evidence types separate. There is no meaningful grand total.

Match each source to the question, then seek customer and branch outcome data.

Limit: 358 review IDs, 95 annotation units and 38 URLs are distinct units, not additive customer counts. The new 16 targeted nonblind AI checks are separate from the first-wave 30 checks; neither is human validation or independent outcome verification.

Counts describe records, annotations and documents. They are not counts of verified customers. Evidence: extension_reviews and frozen reviews tables.

(certain) AUXILIARY METHOD TEST

Two automated label views disagreed on 41.35% of rows.

The largest review appendix covers Indonesian public-service apps, not Digiroom. Its rating-derived and VADER labels disagree on 255,414 of 617,722 aligned rows. Neither is human ground truth. The two label files describe the same base records, with 617,703 distinct composite keys and 19 duplicate excess rows per view. They are useful for testing shortcuts, not for expanding the Auto2000 customer sample.



58.65%

Same label

362,308 aligned rows

41.35%

Different labels

255,414 aligned rows

Keep a star rating, sentiment label, specific task and completed outcome as different fields.

Limit: Historical auxiliary records, May 2012-Mar 2023. Agreement is not accuracy; duplicate label views are not extra people.

IGAR V2,doi: 10.17632/7zryc6k76z.2,CC BY4.0. Paired raw files and row-level nontext comparison are included in the primary ZIP.

(certain) SOURCE-SCOPE AUDIT

Some supplied numbers need narrower denominators.

Several friend-source claims become more useful after checking who answered each question. The older Deloitte charging result uses a small EV-intender subgroup, not all 1,004 Indonesia respondents. PwC's 71% maintenance answer is conditional on owners considering a return to ICE. The supplied Rakuten 57% versus 28% awareness values could not be independently verified from the underlying chart.

Deloitte 2025 charging preference

The 62% home-charging result uses Q43's 178 BEV/PHEV intenders, not all 1,004 Indonesia respondents.

PwC maintenance concern

71% refers to owners considering a return to ICE, not all owners in the study.

Rakuten brand awareness

The supplied 57% versus 28% chart values remain unverified; metadata does not verify the numbers.

Keep the population, year, vehicle definition and question beside every percentage.

Limit: Unverified values remain supplied claims. Conflicting secondary Populix geography and undisclosed subgroup bases remain unresolved.

Friend survey audit: 43claims; Deloitte 2025 SEA Q42/Q43; PwC2024/2025 methods; Rakuten chart metadata versus hidden numeric values.

(certain) EVIDENCE BOUNDARY

The strongest open questions are operational and commercial.

The research found existing functions, specific presentation issues and several plausible opportunities. It did not measure actual branch capacity, fulfilled jobs, current owner entitlement, realized margins or incremental response. A clearer request, record or certificate may help. Which change produces additional value is still a hypothesis.

What is supported

Existing public functions

Specific rendered states

Reproduced calculations

Source and scope corrections

What remains open

Eligible demand and capacity

Fulfilled work and margins

Appraisal acceptance

Incremental customer response

Use the solution menu to choose a question worth resolving, rather than treating page counts as proof of impact.

Limit: No human interviews, branch visits, private account records, submitted bookings or causal Auto2000 experiment were conducted.

First-wave numerical QA: 157 assertions, 140 pass/17 qualified/0fail. Extension: 69 assertions, 66 pass/3 qualified/0fail. AI checks; no human validation.

(certain) RESEARCH STORY

Use this story to navigate the full research.

Read the complete analysis

Workstream reports and 16 cross-source insight cards retain details that are intentionally short here.

Continue in another AI session

context/Auto2000_Master_AI_Context_Updated.md contains the extensive updated case context, evidence capsules, corrections and open questions.

Inspect the underlying data

raw_licensed_data/ contains the original permitted source files. The primary download includes both 617,722-row IGAR label views and the five exact telemetry MAT bundles.

Reproduce the supported calculations

Code, source hashes, datasets, numerical checks and scope notes are included. Some historical live-web recipes still need omitted source caches; this is not a one-click replay of the internet.

Use the correct archive variant

The public backup omits original IGAR review text for privacy. For smaller hosts, the core plus five whole-MAT ZIP volumes extract into the same directory. The main ZIP retains the original licensed files.

Keep the earlier work

Both previous ZIP releases are included unchanged. Their plans and ceilings are historical; the final reports explain what was actually completed.

Style inspiration: the supplied four-page audit PDF, used only for layout. No unrelated case facts or artwork were imported. This is a research reading edition, not a competition submission or implementation plan.